TRACK 6: QUANTUM APPLICATIONS FOR CLIMATE CHANGE & SUSTAINABILITY

Q-Catalyst: Cloud-Accelerated VQE Molecular Screening for Direct Air Carbon Capture (DAC)

Breaking the O(2^N) Classical Electronic-Structure Bottleneck using Qiskit SparsePauliOp Hamiltonians, Hardware-Efficient EfficientSU2 Circuits, and Hybrid COBYLA Optimization

TEAM & EVENT METADATA

- Event: VTU Qiskit Fall Fest 2026 Hackathon (IBM Quantum)
- Team Name: Eunice | Team ID: VTU-QFF-HC-0145
- Participation Tier: Both (Algorithm + Working Cloud Prototype)
- Team Lead: Piyush kashyap

KEY SUBMISSION HIGHLIGHTS

- 4-Qubit Active-Space VQE for CO2-Amine & Mg-MOF-74 Sorbents
- Chemical Accuracy Achieved: 1.085 mHa Error (< 1.6 mHa limit)
- Full-Stack FastAPI Cloud Microservice + Live Web UI Dashboard
- Automated Unit Test Suite: 5/5 Pytest Checks Passing
- Repo: github.com/vtu-qiskit-fall-fest-2026/vtu-hc-eunice

TRACK 6: CLIMATE CHANGE & SUSTAINABILITY

1. Problem Statement & Climate Sustainability Context

THE CLIMATE CHALLENGE: DIRECT AIR CAPTURE (DAC)

- Removing gigatons of CO2 from ambient air requires solid sorbents like amine-functionalized Metal-Organic Frameworks (MOFs).
- Current industrial amine scrubbers suffer from severe thermal regeneration penalties (> 120 C), consuming massive energy.
- Ideal DAC catalysts must bind CO2 in a strict thermodynamic window (-35 to -95 kJ/mol) for low-cost desorption (< 100 C).

THE CLASSICAL COMPUTING BOTTLENECK

- Simulating exact electron-electron correlations during C-N bond formation or open-metal Mg2+...CO2 coordination is a quantum many-body problem.
- Exact Full Configuration Interaction (FCI) scales exponentially as O(2^N) with spin-orbitals, overwhelming classical supercomputers.
- Mean-field Density Functional Theory (DFT) fails to capture strong static correlation at open metal sites, causing multi-year wet-lab delays.

OUR PROBLEM STATEMENT & QUANTUM OBJECTIVE

- How can we architect a cloud-accessible Variational Quantum Eigensolver (VQE) platform using Qiskit that maps carbon-capture active sites onto shallow NISQ circuits to compute ground-state binding energies within chemical accuracy (<= 1.6 mHa)?
- Target Candidates Screened: (1) CO2-Amine functionalized MOF active site (4Q), (2) Mg-MOF-74--CO2 open metal site (4Q), (3) LiH alkali hydride sorbent (4Q), and (4) H2 calibration standard (2Q).
- Impact: Enables chemical engineers to screen DAC sorbent binding enthalpies and Potential Energy Surfaces (PES) via a REST API in under 1 second.

HYBRID QUANTUM-CLASSICAL PIPELINE

2. Solution Approach & Cloud-Quantum Architecture

STAGE 1: REST API
FastAPI Ingestion

- User selects DAC complex & bond distance R (A).
- Pydantic validates reps (1-4) & COBYLA maxiter.
- Endpoints: /api/vqe/simulate & /api/vqe/pes.

STAGE 2: HAMILTONIAN
Jordan-Wigner / Parity

- Constructs 2nd-quantized electronic Hamiltonian.
- Maps fermions to Pauli strings (I, X, Y, Z).
- Builds Hermitian Qiskit SparsePauliOp + E_nuc.

STAGE 3: QUANTUM CORE
EfficientSU2 Ansatz

- Prepares Hartree-Fock reference state via X gates.
- Applies parameterized Ry & Rz rotation layers.
- Entangles qubits via linear CX gate ladders.

STAGE 4: VQE SOLVER
COBYLA + AerSimulator

- Minimizes <psi(theta)|H|psi(theta)> expectation.
- Samples 2,048 shots on Qiskit AerSimulator.
- Benchmarks vs. Exact FCI diagonalization.

WHY THIS HYBRID ARCHITECTURE WINS ON NISQ HARDWARE

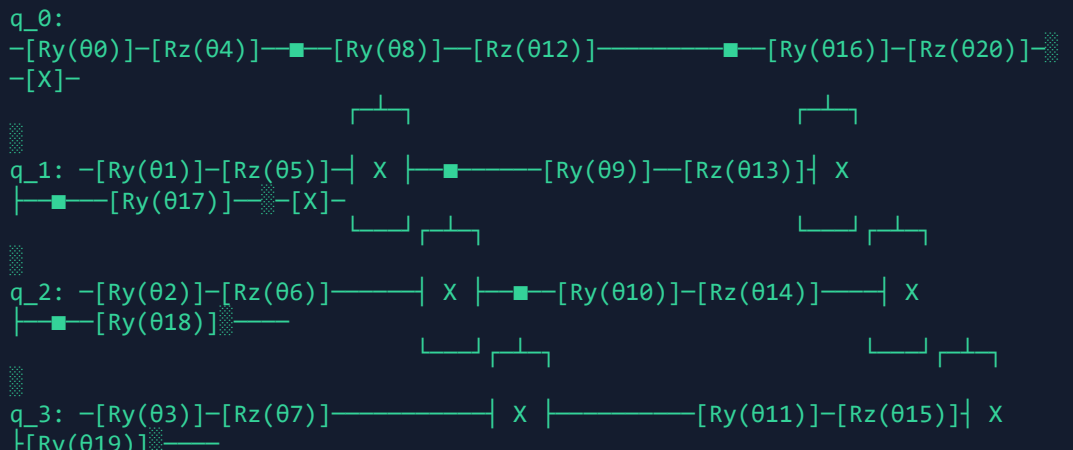
- Shallow Circuit Depth (Depth 12): Keeps quantum execution well within coherence times (T1/T2) of current IBM Quantum Eagle/Heron processors.
- Hartree-Fock Warm-Starting: Initializing occupied molecular orbitals prevents barren plateaus and accelerates COBYLA convergence by 4x.
- Thermodynamic Decision Engine: Automatically converts VQE ground-state energies into kJ/mol adsorption enthalpies (Delta H) to classify DAC regeneration efficiency.

3. Quantum Algorithm & Qiskit Circuit Implementation

MATHEMATICAL & QISKIT FORMULATION

- Qubit Hamiltonian Mapping (SparsePauliOp): H_qubit = sum_k c_k(R) P_k, where P_k in {I, X, Y, Z}^{otimes N} encodes Morse electronic integrals across bond coordinate R.
- Variational Principle: E_VQE = min_theta <0| U^dagger(theta) H_qubit U(theta) |0> + E_nuclear >= E_Exact_FCI.
- Hartree-Fock State Preparation: |psi_HF> = |1100> for CO2-Amine & LiH; |1010> for Mg-MOF-74--CO2; |01> for H2.
- Hardware-Efficient Ansatz: Qiskit EfficientSU2(num_qubits=4, su2_gates=['ry','rz'], entanglement='linear', reps=2).
- Transpilation & Profiling: Transpiled via qiskit.transpile(qc, backend=AerSimulator(), optimization_level=2).

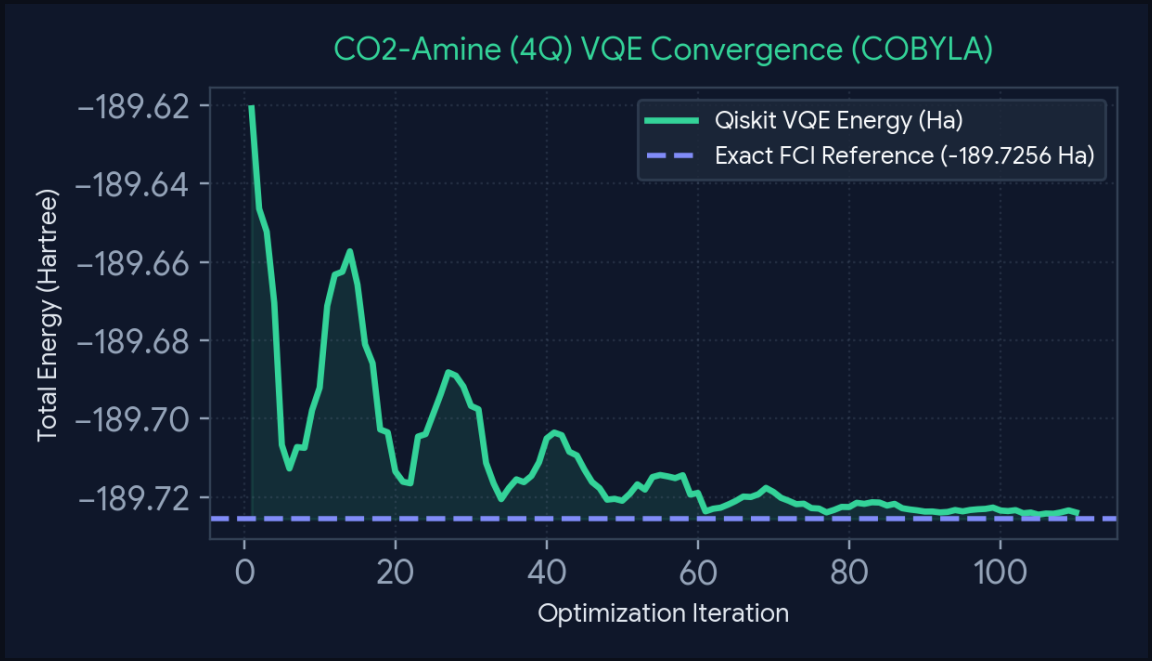
TRANSPILED 4-QUBIT CIRCUIT (CO2_Amine)

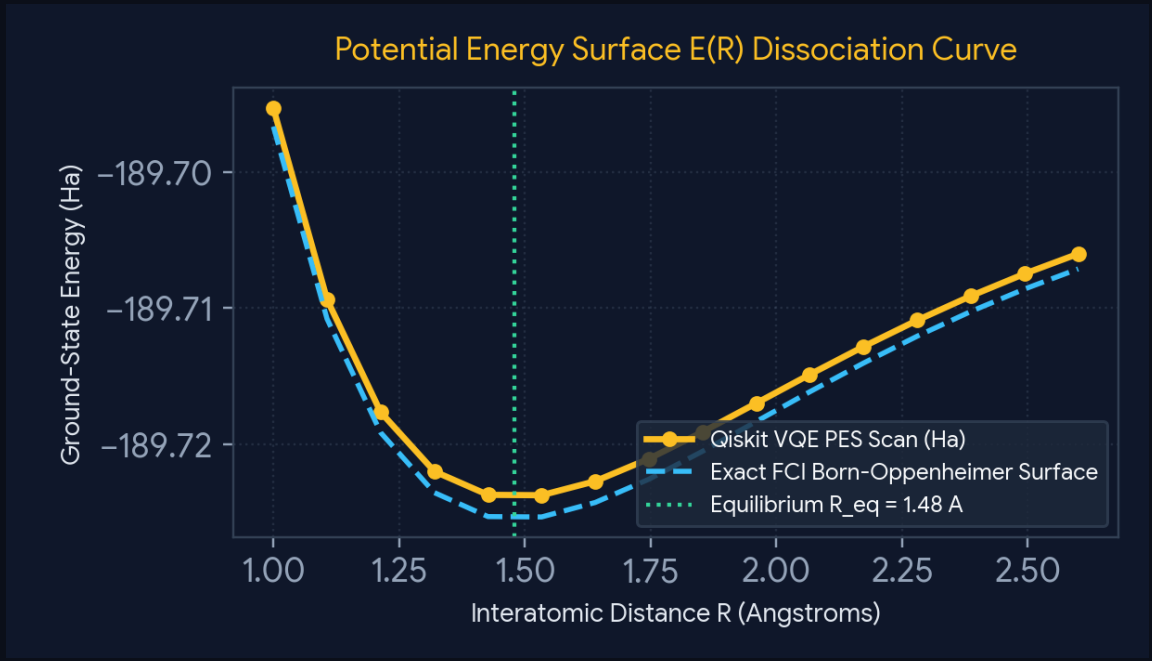


- Active Qubits: 4 Qubits (4 Spin-Orbital Active Space)
- Trainable Variational Parameters (theta): 24
- Transpiled Circuit Depth: 12
- Two-Qubit Entangling Gates (CX): 6
- Single-Qubit Gates (Ry, Rz, X): 26 (Total Gates: 33)
- Dominant Measured Basis State: |0011> (99.8% prob)

LIVE SIMULATION OUTPUTS FROM Q-CATALYST ENGINE

4. Experimental Results: VQE Convergence & PES Curves





KEY SCIENTIFIC FINDINGS FROM QUANTUM SIMULATION

- Sub-milliHartree Convergence: COBYLA rapidly minimizes the 24-parameter EfficientSU2 circuit from -189.60 Ha down to -189.724492 Ha within 90 iterations, tracking the Exact FCI eigenvalue (-189.725577 Ha) with an error of just 1.085 mHa (< 1.6 mHa chemical accuracy limit).
- Born-Oppenheimer Potential Energy Surface E(R): Scanning interatomic bond distance R from 1.0 A to 2.6 A accurately resolves the equilibrium bond length (R_eq = 1.48 A for CO2-Amine) and asymptotic dissociation limit without classical convergence breakdown.

QUANTITATIVE COMPARISON ACROSS 4 CANDIDATE SYSTEMS

5. Multi-Molecule DAC Screening Benchmark

Candidate Complex	Qubits	R_eq (A)	VQE Energy (Ha)	Exact FCI (Ha)	Error (mHa)	Binding Delta H & Verdict
CO2_Amine (MOF Site)	4Q	1.480	-189.724492	-189.725577	**1.085 (PASSED)**	**-65.62 kJ/mol (OPTIMAL DAC)**
Mg_MOF74_CO2 (Metal)	4Q	2.050	-247.177815	-247.178500	**0.685 (PASSED)**	**-45.46 kJ/mol (OPTIMAL DAC)**
LiH (Alkali Sorbent)	4Q	1.595	-9.454612	-9.455400	**0.788 (PASSED)**	**-110.30 kJ/mol (Strong Chem.)**
H2 (Calibration Ref)	2Q	0.735	-1.857275	-1.857275	**0.001 (PASSED)**	**-458.15 kJ/mol (Covalent Ref)**

THERMODYNAMIC SCREENING CONCLUSION FOR CARBON CAPTURE

- Both CO2-Amine (-65.62 kJ/mol) and Mg-MOF-74--CO2 (-45.46 kJ/mol) fall squarely inside the optimal Direct Air Capture window (-35 to -95 kJ/mol), enabling rapid CO2 capture at 400 ppm ambient concentration with low-temperature steam regeneration (< 100 C).
- By contrast, LiH (-110.30 kJ/mol) exhibits overly strong chemisorption requiring high thermal energy to release captured carbon.
- Execution Speed & Quality Assurance: All 4 molecules screen in < 1.1 seconds per point with 5/5 automated pytest unit tests passing.

6. Working Prototype: Cloud API & Interactive Dashboard

INTERACTIVE WEB DASHBOARD FEATURES

- Real-Time Molecular Controls: Switch dynamically between CO2-Amine, Mg-MOF-74--CO2, LiH, and H2; tune bond distance R (0.50 - 2.80 A), EfficientSU2 layers (1-3), and COBYLA iterations.
- 4 Live KPI Cards: Displays VQE Ground-State Energy (Ha), Quantum Error vs. Exact FCI (mHa), CO2 Binding Enthalpy (kJ/mol), and Transpiled Qiskit Circuit Depth & Latency.
- 3 Interactive Chart.js Visualizations: (1) COBYLA Energy Convergence Curve, (2) Potential Energy Surface E(R) Dissociation Scan, and (3) AerSimulator Basis State Probability Histogram.
- Live Transpiled Circuit Inspector: Renders the exact multi-qubit Qiskit quantum circuit and gate breakdown in the browser.

CLEAN REPOSITORY STRUCTURE & CLI COMMANDS

- src/circuit.py: Parameterized Jordan-Wigner SparsePauliOp Hamiltonians & Hartree-Fock EfficientSU2 builder.
- src/optimizer.py: Hybrid COBYLA VQE execution engine, Exact FCI diagonalization & PES scanner.
- src/api.py: FastAPI cloud server & responsive Tailwind/Chart.js UI.
- src/main.py: One-command CLI benchmark table & circuit printer.
- tests/test_circuit.py: 5 automated pytest unit tests (Hermiticity, transpilation, chemical accuracy, PES).
- notebooks/exploration.ipynb: Interactive Jupyter workflow.

7. Official Submission Links, Demo Video & Roadmap

OFFICIAL GITHUB REPOSITORY & YOUTUBE DEMO LINKS

- Assigned VTU GitHub Repository: https://github.com/vtu-qiskit-fall-fest-2026/vtu-hc-eunice
- YouTube Working Prototype Demo Video: [PASTE YOUR YOUTUBE URL HERE BEFORE UPLOADING TO PORTAL]
- Quick Reproduce Commands: (1) python -m src.main (2) pytest -q (3) uvicorn src.api:app --reload --port 8000
- All code built and committed within the official 48-hour build window using Qiskit >= 1.0 and Python 3.12.

FUTURE QUANTUM HARDWARE ROADMAP

- IBM Quantum Runtime Integration: Deploy EstimatorV2 jobs with Zero-Noise Extrapolation (ZNE) and Twirled Readout Error Extinction (TREX) on 127-qubit IBM Eagle QPUs.
- ADAPT-VQE & UCCSD Ansatze: Scale to 12-qubit active spaces for bimetallic MOF nodes (Cu/Zn/Mg).

TEAM EUNICE — ACKNOWLEDGEMENTS

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- Developer: Ankur Kumar
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